

Alexie Wyant Clover

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PROFESSIONAL SUMMARY

Results-driven Chemist with successful experience in designing, executing, and optimizing complex methods. Proficient in instrumental analysis techniques, method development, and validation, consistently delivering accurate and actionable data. Adept at troubleshooting and resolving technical challenges to maintain laboratory efficiency. Strong leadership skills demonstrated through successful mentorship of junior chemists and cross-functional collaboration.

Education

Western Washington University (WWU)	2021-2023
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Master of Science, Organic Chemistry

- Thesis: New Synthetic Methods Based on Silicon-Tethered Nucleophilic Addition Reactions
- Teaching and Learning Certificate

Northern Arizona University (NAU)	2016-2021
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Bachelor of Science in Chemistry

Bachelor of Science in Biology

- Achieved the Nancy & Henry Wettaw Organic Chemistry Award in 2020.

Curtin University	2018
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Studied Abroad in Western Australia, from July 2018 to December 2018

Publications / Presentations

- "Optimization of Reaction Conditions for Oligo-peptide Conjugation Reactions" Poster ADC World Conference 2024
- Clover, A. W., Jones, A. P., Berger, R. F., Kaminsky, W., & O'Neil, Gregory. W. (2024). Regioselective fluorohydrin synthesis from allylsilanes and evidence for a silicon-fluorine gauche effect. *The Journal of Organic Chemistry*, 89(7), 4309–4318. <https://doi.org/10.1021/acs.joc.3c02163>
- "Regioselective Fluorohydrin Synthesis from Allylsilanes" Poster ACS Conference 2023

Work Experience

LGC Axolabs Product Development Chemist:	2023-Current
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- Optimize and scale up/down solid-phase oligonucleotide synthesis, purification, conjugation, and downstream processing methods in preparation for handover to routine manufacturing.
- Provide technical and early-stage manufacturing support in the development of therapeutic GMP manufacturing process and capability, including developing and drafting of manufacturing procedures.
- Provide technical support in root cause failure analysis for production and quality incidents.
- Keep good records of production activities, experimental procedures, and analytical data.
- Maintain the chemistry laboratory space in compliance with safety regulations.
- Provide technical customer support and routine progress updates to clients in conjunction with the project manager assigned to each specific project.

R &D Organic Chemist WWU:**2021-2023**

- Experience in complex organic synthesis and running analytical procedures.
- Experience with root cause failure analysis. Working in a small lab with limited resources required excellent troubleshooting skills along with the ability to fix things when they break.
- Experience working with strong acids and bases. Including HF acid. Along with experience working in glove boxes.
- Isolation and classification of organic and inorganic compounds through liquid column chromatography and crystallography.
- Spectroscopy, including nuclear magnetic resonance (NMR) (1D & 2D), IR, and MS.

Teaching Assistant WWU:**2021-2023**

- Oversaw various laboratory sections encompassing both online and hands-on experimental components.
- Worked across a wide array of chemistry domains, including General Chemistry I/II, Physical Chemistry I/II, Inorganic Chemistry I/II, and Organic Spectroscopy.
- Managed an independent laboratory and provided valuable assistance to professors in conducting advanced experiments.
- Developed effective experimental protocols, ensured rigorous laboratory safety, and provided clear explanations of complex chemical reactions.

Supplemental Instructor NAU:**2019-2021**

- Led Organic Chemistry II as a skilled Supplemental Instructor, conducting engaging discussion and problem-solving sessions.
- Initiated and managed personalized classroom study sessions, curating original content to facilitate students' comprehension of organic chemistry concepts.
- Collaborated closely with professors to develop practice exams that aligned with course objectives and requirements.

Private Tutor:**2019-2021**

- Provided comprehensive tutoring for both Organic Chemistry I and II, delivering sessions through both Zoom and in-person formats.

Skills and Proficiencies

- Expertise in utilizing advanced software for data analysis and interpretation.
- Exceptional attention to detail, combined with a track record of ensuring laboratory safety protocols.
- Experience using Unicorn software and method writing.
- Proven ability to troubleshoot instruments.